

Multiple Choice

1. **Answer:** D

Options: A) Nitrogen; B) Argon; C) Methane; D) CO₂

Note: The correct answer is CO₂ because the Martian atmosphere is composed of approximately 95% carbon dioxide, making it inhospitable for human respiration, which requires oxygen.

2. **Answer:** C

Options: A) A kilogram of feathers; B) Five pounds of bricks as measured on Earth; C) Five kilograms of feathers; D) A kilogram of bricks

Note: Five kilograms of feathers weighs the most on the moon because weight is determined by mass and gravitational pull, and since the mass remains the same regardless of location, five kilograms will always weigh five kilograms, even in the lower gravity of the moon.

3. **Answer:** C

Options: A) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.; B) After you've repeated your experiment many times.; C) You can never prove your theory to be correct only "yet to be proven wrong".; D) When you and many others have tested the hypothesis.

Note: The correct answer reflects the philosophical principle of scientific theories, which states that theories can only be falsified through empirical testing, rather than definitively proven true, as new evidence may always emerge that contradicts the existing theory.

4. **Answer:** B

Options: A) 130 million km; B) 150 million km; C) 170 million km; D) 190 million km

Note: One astronomical unit (AU) is defined as the average distance from the Earth to the Sun, which is approximately 150 million kilometers.

5. **Answer:** C

Options: A) 240 to 680 nm.; B) 360 to 620 nm.; C) 380 to 740 nm.; D) 420 to 810 nm.

Note: The visible part of the electromagnetic spectrum ranges from approximately 380 to 740 nanometers, which corresponds to the wavelengths of light that can be detected by the human eye, allowing for the perception of color.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The Tunguska event, which occurred in 1908, was caused by an airburst of a

meteoroid or comet and is unrelated to the mass extinction event that wiped out the dinosaurs 65 million years ago, which is attributed to a separate impact event likely involving a large asteroid.

7. **Answer:** False

Options: A) True; B) False

Note: The United States has never successfully sent landers to Venus that operated on its surface and transmitted data back to Earth, as all American missions to Venus, such as the Venera program, were conducted by the Soviet Union.

8. **Answer:** False

Options: A) True; B) False

Note: Titan's atmosphere is predominantly composed of nitrogen and methane, with only trace amounts of oxygen, distinguishing it significantly from Earth's nitrogen-oxygen rich atmosphere.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because Mercury's elliptical orbit causes it to be approximately 1.5 times closer to the Sun at perihelion than at aphelion, resulting in a solar radiation flux that is about 2.25 (or $9/4$) times greater due to the inverse square law of radiation.

10. **Answer:** True

Options: A) True; B) False

Note: A day on the Moon, known as a lunar day, lasts about 29.5 Earth days because the Moon takes this amount of time to rotate once on its axis relative to the Sun, which is due to its synchronous rotation with Earth.

Long Form Response

11. **Answer:** Formation theories of our solar system must address several essential characteristics, including the initial conditions of the solar nebula, the processes of accretion and differentiation, and the dynamical interactions among celestial bodies that led to the current structure of the solar system. These theories, such as the nebular hypothesis, focus primarily on the physical and chemical processes that govern the formation of planets and other bodies, without delving into biological factors. Consequently, the presence of life on Earth is not included in these explanations because formation theories are concerned with the abiotic origins of solar system components rather than the subsequent development of life, which is a distinct area of study within astrobiology and planetary science. This separation underscores the complexity of life's emergence as a phenomenon that requires a different analytical framework, one that incorporates biological, environmental, and evolutionary factors beyond the scope of formation theories.

Note: correct because it identifies the critical aspects of solar system formation theories, including initial conditions, accretion processes, and dynamical interactions, while

appropriately noting that these theories focus on physical and chemical processes rather than biological factors like the presence of life on Earth.

12. **Answer:** To determine the distance to the rock identified as "bigfoot" on Mars, scientists utilized its angular size and assumed physical dimensions. By measuring the angular size of the rock from the rover's camera and knowing its approximate height, calculations were performed using the formula: $\text{distance} = \text{size} / (2 * \tan(\text{angular size} / 2))$. Assuming "bigfoot" has a height of around 6 meters, the angular measurement translated to a distance of approximately 6 meters from the rover. This methodology not only aids in understanding the rock's physical scale but also informs further rover observations by establishing a baseline for similar features in the Martian landscape. Such measurements are crucial for interpreting geological processes and assessing the potential for past life on Mars.

Note: correct because it accurately applies trigonometric principles to calculate the distance to the rock based on its angular size and assumed physical dimensions, demonstrating a clear understanding of how observations from a rover can be quantitatively analyzed in the context of planetary science.